20

Anticoagulation and Bleeding Management

When I am consenting a family for extracorporeal membrane oxygenation (ECMO) and discussing the risks and benefits to initiating support, bleeding and clotting are the most common risks that are brought up, second only to the procedure itself. When we consider ECMO for patients, we are weighing the benefits against the risks, chief of which are bleeding and thrombosis.

This is not surprising. Bleeding is the most common complication reported in ECMO, occurring in as many as half of all cases, with clotting/thrombosis not far behind, occurring in a quarter to a third of all cases. While bleeding/thrombosis are common, we can equip ourselves with a much better ability to manage and mitigate these complications by developing an understanding of the mechanisms by which we are able to anticoagulate the blood and how those lead to bleeding in patients on ECMO.

WHY DO WE NEED TO ANTICOAGULATE IN ECMO?

Normally, blood is pumped from the heart, through a continuous network of blood vessels all comprised of tissue that the body and immune system recognizes. So what happens when blood is pumped through the ECMO circuit? Consider the differences:

- It is sucked through a negative venous drainage cannula.
- It takes large and small turns introducing turbulence.
- It is exposed to a motor turning at 1500–10,000 revolutions per minute with shear stress.
- It is exposed to a host of plastics and other foreign substances, which activate the immune and complement systems.
- It is cooled when it goes out of the body and rewarmed by the water from the heater cooler.

The take-home point being that there is nothing natural or normal about extracorporeal blood flow. The result is that the blood flowing from the extracorporeal circuit has a higher likelihood of being inflamed and is more likely to clot. Hopefully as the technology for circuits/pumps continues to improve, this trauma and inflammatory activation of blood will be minimized. However, in the interim, the solution is management of the thrombosis effect through anticoagulation.

UNDERSTANDING THE PRINCIPLES BEHIND HEMOSTASIS

Many practitioners struggle with anticoagulation and bleeding management with good reason – the way the body forms and breaks down clot is a dynamic and complex system. That said, there are some principles that we can establish which hopefully will give a framework for approaching the management of anticoagulation on ECMO. The best way to build this framework is to start by examining how bleeding is stopped by the body on a basic level. Let's review.

Let's say you are preparing dinner – and tonight is kebab night! However, as you are slicing the peppers, your mind drifts and you accidentally cut your finger. As you rush to hold pressure, your mind again drifts to imagining how your body can stop this bleeding.

Bleeding is occurring in this case due to the disruption in the integrity of the blood vessel due to a brutally sharp kitchen knife (Fig. 20.1).

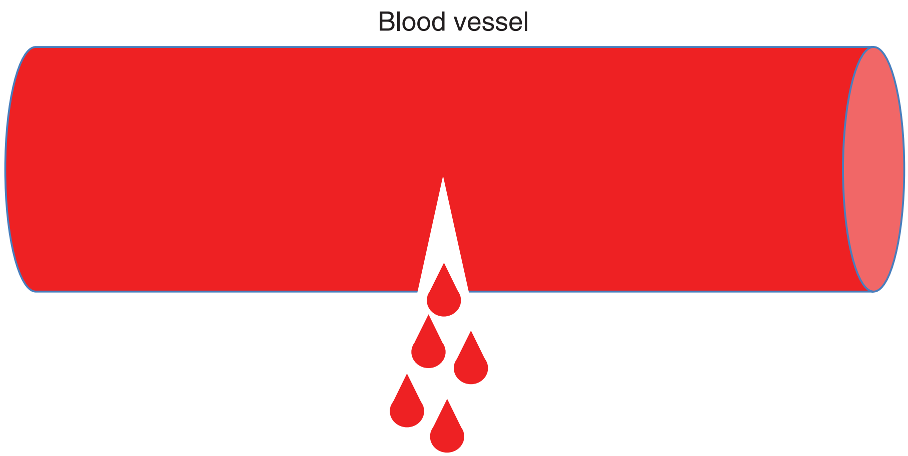

FIG. 20.1 Bleeding due to disruption of the blood vessel

At a basic level, your body is able to stop the bleeding through two mechanisms: creation of an initial web/scaffolding (fibrin) and adherence of cells to plug the hole (platelets). Clearly, both elements are necessary – without the web, the platelets would have nothing to adhere to, and without the platelets, there would be no integrity to whatever initial scaffolding is laid down (Fig. 20.2).

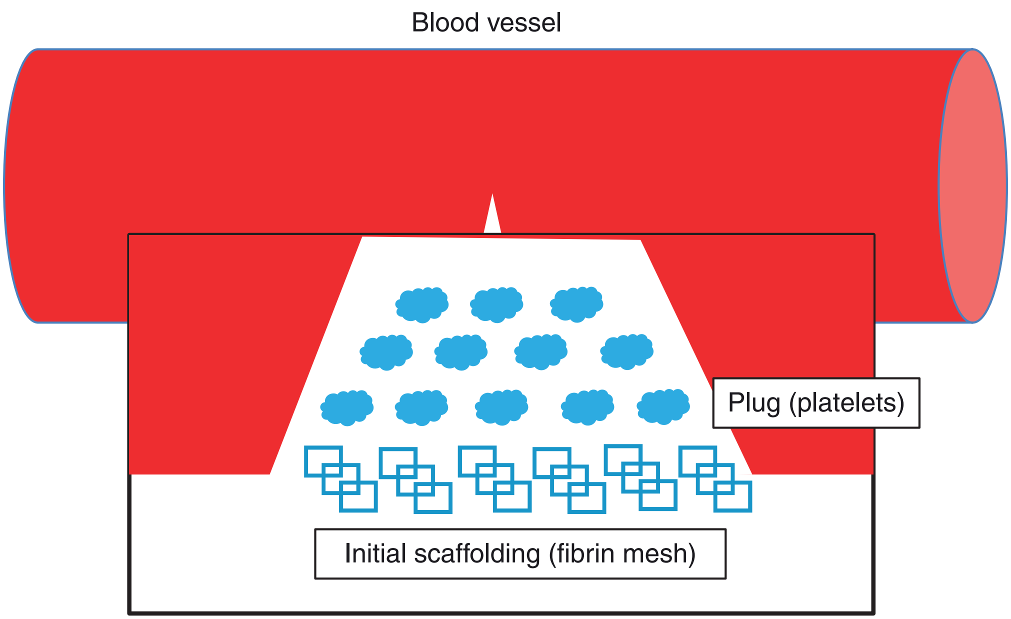

FIG. 20.2 Clotting at the source of bleeding with fibrin mesh and platelet plug

The clot then can become stronger as it progresses from three phases:

1. **Initiation:** Initial mesh generation at the site of injury
2. **Propagation:** Platelet adhesion to the fibrin mesh, activation of factors needed to solidify fibrin mesh
3. **Amplification:** Platelets aggregating to each other with fibrin mesh maturation

You can quickly imagine why this requires a dynamic process – you need to form clot when there is bleeding and not form clot when there is no bleeding. Any breakdown in the system can lead to excessive thrombosis or excessive bleeding. The body is able to regulate this through the coagulation cascade.

THE COAGULATION CASCADE: HOW CLOTTING IS REGULATED

There are many factors that go into the coagulation cascade, so let's break things down into smaller steps. Ideally, you want to accomplish the following steps:

Step 1: Form clot where there is bleeding
Step 2: Lay down initial framework for clot
Step 3: Activate platelets to form the plug
Step 4: Form enough clot to stop bleeding
Step 5: Stop clotting when bleeding stops
Step 6: Break down clot

Let's break down each step and the factors that go into each.

Step 1: Form Clot Where There Is Bleeding

You need the initiation of this cascade to be specific to the spot where there is bleeding. Otherwise, clotting can occur anywhere, and would not focus on places where the integrity of the blood vessel has been disrupted, like our unfortunate lacerated finger, for example.

The way this happens is that the blood vessel that is damaged releases tissue factor. This combines and activates factor VII. You may recognize activated factor VII, as this is what is sometimes administered in severely bleeding patients. This is one of the reasons it is administered – it is the initial entrance point into the coagulation cascade (Fig. 20.3).

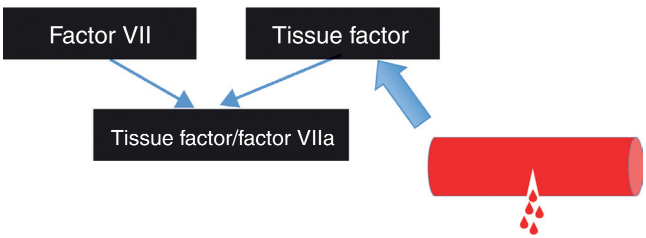

FIG. 20.3 Initial activation of the coagulation cascade through the release of tissue factor

Step 2: Lay Down Initial Framework for Clot

Remember that the scaffolding of our clot is fibrin mesh – it is what the platelets adhere to. The cascade that activates fibrin and allows for this fibrin mesh generation is initiated by the activated factor VII/tissue factor complex. This does not happen directly but rather happens through the activation of other factors: **tissue factor/factor VIIa** activates **factor X**, which activates **thrombin**, which activates **fibrinogen** (Fig. 20.4).

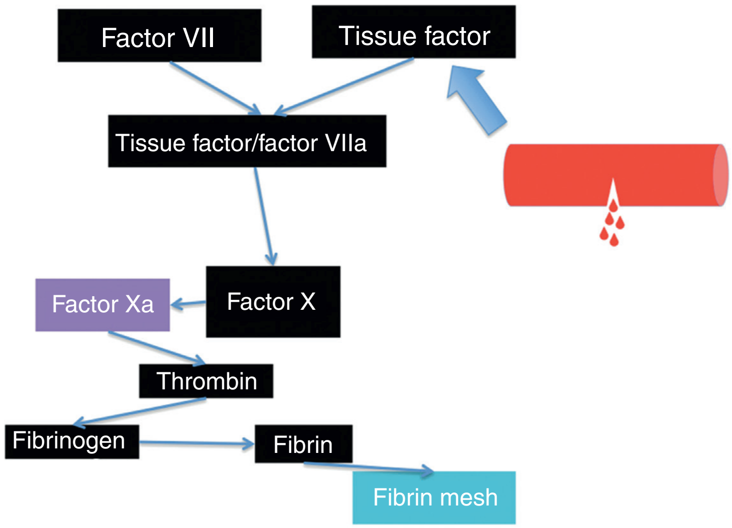

FIG. 20.4 Activation of the initial framework for the clot

Why all these steps? Every extra step represents a point where the cascade can be regulated, which is a good thing. An unregulated system ultimately leads to a disorder of bleeding or clotting. Rather, this becomes a delicately arranged symphony, allowing for the perfect balance. Pay attention to the factors mentioned; they are not arbitrary and will ultimately play a specific role in how we manage our patients.

Step 3: Activate Platelets to Form the Plug

Now that we are done with step 2, we have our scaffolding started and are ready to start to form the plug! Remember, our plug is made of platelets.

Platelets need to be activated by our cascade so that they are ready to adhere to the fibrin mesh only when there is bleeding. Ideally, the factor that activates these platelets would be downstream of the cascade, capturing the feedback of the entire cascade, without being the end product of the cascade. That is exactly what happens with thrombin, our second to last factor that activates platelets! Pay attention to thrombin, this is not the last time it will play a role in our clotting cascade (Fig. 20.5).

Step 4: Form Enough Clot to Stop Bleeding

Now we have our scaffolding (fibrin mesh) set up at the site of bleeding and have started to plug the hole with platelets. All that is left to do is to ramp up the system, increasing the activation of platelets and the generation of fibrin mesh.

Doing this employs another cascade of factors that is intrinsic to the body, rather than extrinsically activated (by tissue factor), and therefore is designated as the *intrinsic* cascade. The steps of the intrinsic cascade (factor IX activating factor XI, which activates factor X) are less relevant to our discussion but what is important is the factor that activated it. You guessed it – our friend thrombin (Fig. 20.6)!

We should pause here and reflect on how thrombin is the lynch pin of this whole process in many ways – this one factor activates our platelets, contributes to the fibrin mesh generation as the precursor to fibrin, and ramps up the system by activating the intrinsic cascade. We will revisit the essential role of thrombin as it plays a key role the regulation of this cascade.

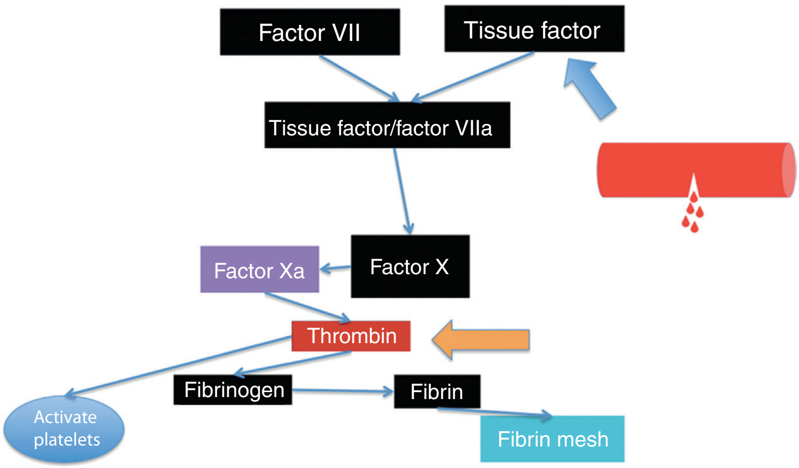

FIG. 20.5 Thrombin activation of platelets

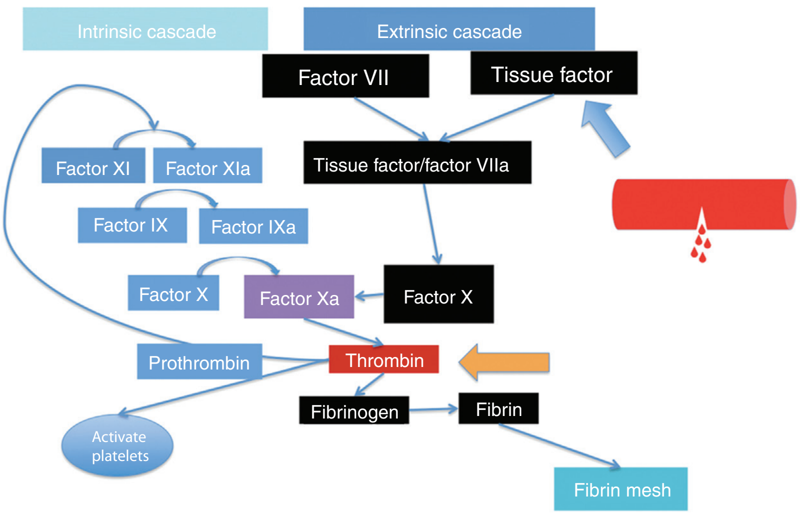

FIG. 20.6 Escalation of the coagulation cascade through activation of the intrinsic pathway

Step 5: Stop Clotting When Bleeding Stops

Congratulations, you now have a strong clot with both activated platelets and a mature fibrin mesh right where you need it, at the site of the injury. You can now release pressure on your bleeding finger and it will not continue bleeding. But the process is not over here. Much like a full bathtub requires that you must turn off the water, the next step is turning off this process. If the system did not have this step, then clotting would continue and excessive thrombosis would ensue (Fig. 20.7).

The system to inhibit clotting is as complex as the system to lay down the clot, with many proteins and enzymes inhibiting factors at different points in the cascade, protein C, protein S, tissue factor pathway inhibitor, and antithrombin to name a few. Subtlety is the name of the game here – multiple enzymes allow for nuance in how the system is broken down.

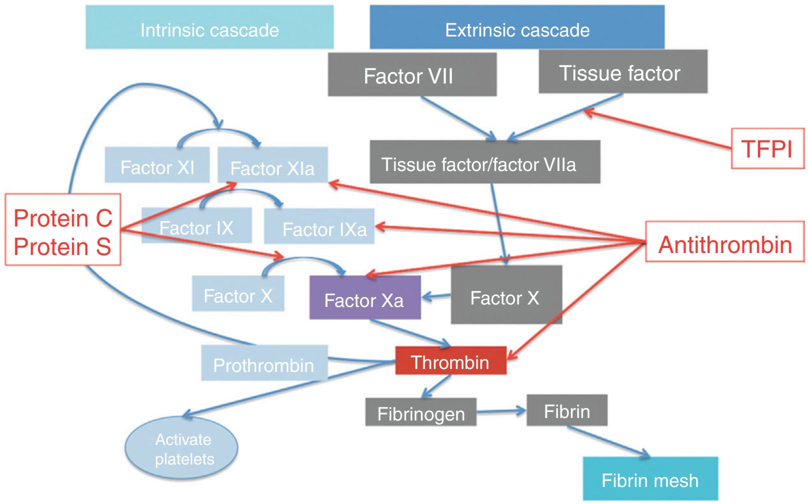

FIG. 20.7 Regulation of the coagulation cascade. TFPI, Tissue factor pathway inhibitor.

Step 6: Break Down Clot

Finally, once the tissue is repaired, there must be a system to actually break down the clot/fibrin mesh. This is different than Step 5 – it is draining the bathtub rather than simply turning the water off, to use our analogy. The process of breaking down our fibrin mesh is referred to as fibrinolysis, and is facilitated by the activation of plasminogen to plasmin, which in turn breaks down the fibrin mesh (Fig. 20.8).

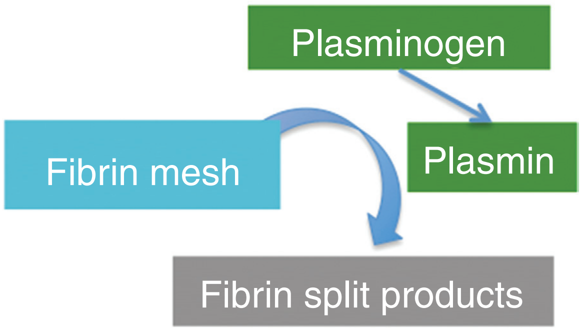

FIG. 20.8 Fibrinolysis through plasminogen activation

ANTICOAGULATION AND ECMO

Even though there was a lot here, we now have a framework that we can apply to better understand the way we use anticoagulation on ECMO. Remember that due to all of the changes described in the beginning of this chapter (turbulence, temperature changes, contact of foreign surfaces, etc.) the coagulation cascade is ramped up. All of the factors that we have described are activated and expressed and the result is that patients are likely to clot. Therefore, we need to employ a method of inhibiting this activation. We do not necessarily want to break up clots, just to stop them from forming.

If we wanted to inhibit one factor that would really disrupt the whole clotting cascade, which one would it be? What about the one factor that is responsible for activating platelets, ramping up the coagulation cascade by activating the intrinsic cascade, and forming the fibrin mesh as a direct precursor to fibrin? That's right – thrombin is our target!

In the majority of cases, thrombin inhibition is accomplished with heparin. As you will recall, the body has a natural mechanism of thrombin inhibition, through employment of the protein antithrombin. Heparin makes use of antithrombin by binding to naturally circulating antithrombin and increasing its effectiveness. Heparin that is bound to antithrombin increases thrombin inhibition 1000-fold (Fig. 20.9).

Heparin is not the only agent that can inhibit thrombin, but since it is the most common, let's explore a little more about how heparin can be used in ECMO.

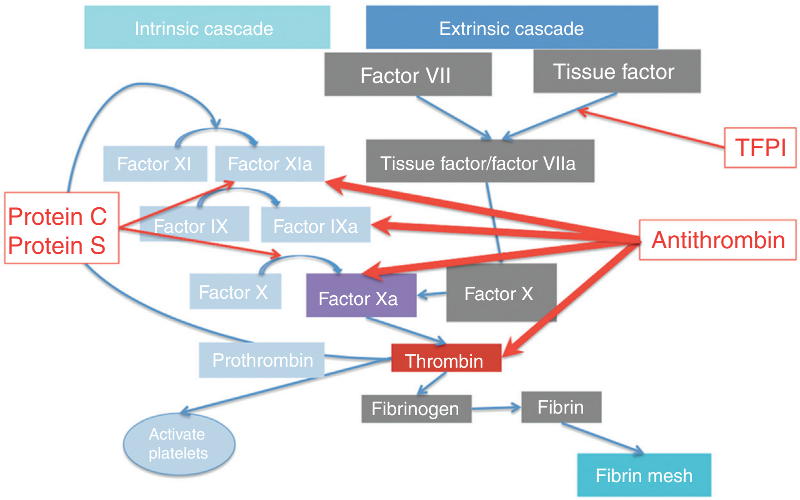

FIG. 20.9 Inhibitory effect of antithrombin. TFPI, Tissue factor pathway inhibitor.

HEPARIN USE IN ECMO

There exists a good deal of variation in how heparin is used in ECMO. We will try to limit our discussion to general concepts, but will give some specifics where helpful. Processes and protocols should be tailored to the needs of individual programs.

There are two ways that heparin can be dosed: **maintenance dose** and **bolus dose** during cannulation. The purpose of the bolus dose is to prevent clots from forming on the wires/cannulas while the cannulation is being completed. It is usually administered after access is completed and before dilation begins. The amount bolused differs but is usually around 50–100 units/kg. Whenever possible, address baseline abnormalities in clotting ability (such as thrombocytopenia, low international normalized ratio [INR], etc.) beforehand so as not to predispose to bleeding during/after cannulation.

Maintenance dose is used to allow for a baseline level of anticoagulation, with the rationale of preventing clots in the circuit, maintain the oxygenator, and prevent clots from happening in the body such as around the cannula or in the left ventricle. Heparin use as maintenance is usually titrated to effect with the aim of maintaining enough anticoagulation to achieve therapeutic effect, without overshooting and predisposing to hemorrhage.

Since the maintenance dose is initiated after the bolus dose is administered, there may be a period where the effect of the bolus dose is wearing off. It will usually take a couple of hours for coagulation parameters to enter into a therapeutic range, signifying that it is time to initiate heparin.

The amount of heparin required differs between patients and titration patterns vary. One approach is as follows: initiate heparin 2 hours after bolus if coagulation parameters are in therapeutic range and titrate every 6 hours based on labs. We will cover labs for measuring effect of heparin later. Bolus doses of heparin during maintenance dosing can lead to overshooting – the risks and benefits should be considered by the treatment team.

Whenever possible, administer heparin to the circuit, attaching to the pre-membrane port if available.

MEASURING THE EFFECT OF HEPARIN

Like most interventions that we have covered to this point, anticoagulation requires diligent observation of the effect, adjusting as necessary. Let's examine the most common ways that we can measure the anticoagulation effect of heparin.

Activated Clotting Time (ACT)

ACT is a commonly used lab tast, particularly in the catheterization lab/operating room. It activates the clotting cascade by adding particulates and then measures the amount of time to clot. The result is a very quick assessment of how quickly the blood is clotting and the degree to which the coagulation cascade is inhibited. The quick/point-of-care nature of this test makes it very useful during cannulation or when an immediate assessment is needed.

However, there are shortcomings of this test. There may be variation between tests and instruments that are hard to correlate clinically, especially in making decisions about the small titrations that need to be done during maintenance dosing of heparin. Additionally, the test only measures the length of time, not anything specific to the coagulation cascade, making it less reliable for understanding the individual effects.

The take-home point being that ACT is a great test for quick assessment and for higher doses of heparin, when heparin effect is the predominant reason for prolongation of the clotting time. However, at the lower maintenance doses used at the bedside for ECMO, the parameters become less useful and are less likely to be affected by the heparin dose administered.

Partial Thromboplastin Time (PTT)

PTT is one of the two parameters that are often measured to assess the ability of the clotting cascade to form clot, along with prothrombin time (PT). Both measure the time to clot, but with different activators added to measure the relative effect of different factors in the coagulation cascade. For PTT, the activators act on the factors of the common and intrinsic cascades while for PT, the activators act on the common and extrinsic cascade. Since antithrombin (and heparin by extension) inhibits the factors of the intrinsic cascade, PTT will correlate with the effect of heparin. The higher the heparin dose, the more effective the activity of antithrombin, and thus the longer the PTT (Fig. 20.10).

Targets for PTT can differ as can the acceptable range. The narrower the range, the closer the anticipated effect of heparin. However, this can become increasingly difficult to accomplish clinically. In general, local protocols should dictate the ideal range for titrating heparin, such as 40–60 seconds for patients at high risk of bleeding or 60–80 seconds for patients with a higher risk of clotting.

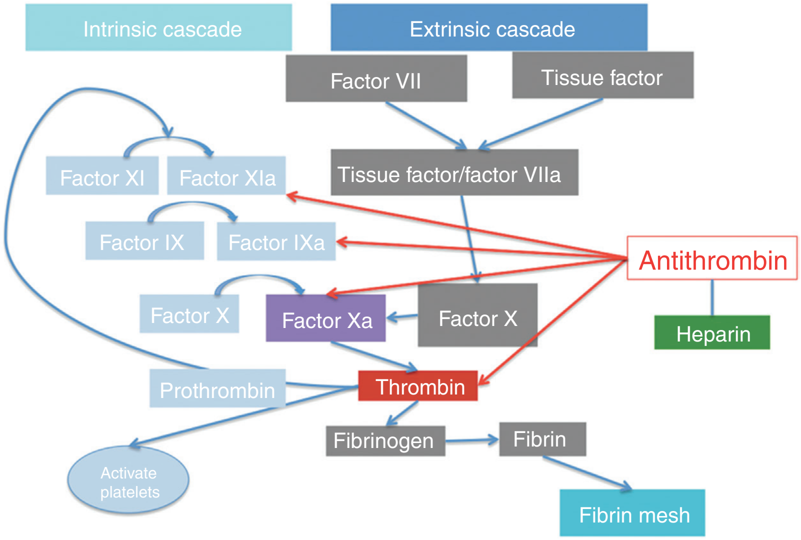

FIG. 20.10 Effect of antithrombin on the factors of the intrinsic pathway

Anti-Xa Levels

A final test often used for measuring the effect of heparin is anti-Xa level. The test involves adding antibodies to factor Xa. Since antithrombin inhibits factor Xa, heparin leads to higher activity of antithrombin, greater inhibition of factor Xa, and higher levels of antibodies to factor Xa. The higher the anti-Xa level, the higher the effect of heparin. Reasonable clinical targets are 0.1–0.3 units/mL for patients at a high risk of bleeding and 0.3–0.7 units/mL for patients with a higher risk of clotting (Fig. 20.11).

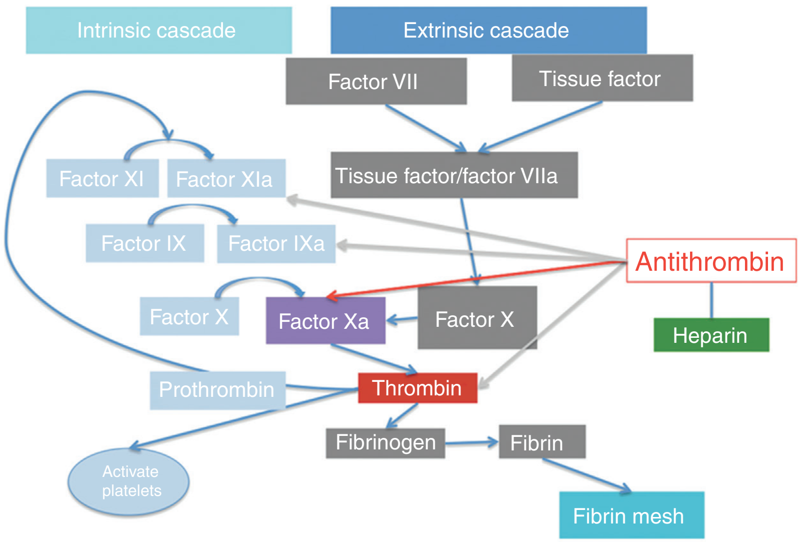

FIG. 20.11 Antithrombin and factor Xa

Since anti-Xa levels are specific to one factor, they tend to be more reliably correlated with heparin effect, rather than other causes of clotting abnormality. However, the availability of the test and speed at which it can be obtained from the lab can vary and may be a limitation to regular use of this test.

Comparing Tests of the Effect of Heparin

If all tests are equally available and cost effective, there can be difficulty reconciling which test to order. While anti-Xa is more specific to the effect of heparin, PTT can be elevated due to a variety of conditions causing depletion of factors common to the intrinsic cascade such as liver disease, disseminated intravascular coagulation, etc.

One strategy can involve intermittently correlating PTT and anti-Xa levels to better understand coagulation status. Anti-Xa will give a more reliable estimate of the effect of heparin, while PTT may be used to ensure that other causes of coagulopathy are not making the anticoagulation effect of heparin dangerously high. As a general rule, the aim should be a consistent and reproducible process with which the whole team is comfortable.

CHALLENGES IN THE USE OF HEPARIN ON ECMO

While there are many advantages to heparin, such as ease of use, familiarity, and cost, there may be challenges associated with its use that should be considered. These include difficulty maintaining a therapeutic window, heparin resistance, and thrombocytopenia. These challenges may lead to consideration of other agents for anticoagulation.

Thrombocytopenia

One of the most common concerns of heparin is the presence of thrombocytopenia, particularly heparin-induced thrombocytopenia, or HIT.

HIT refers to the immune-mediated destruction of platelets that is induced by the presence of heparin. The mechanism of HIT involves the release of platelet factor 4 (PF4), a small, prothrombotic granule that is normally stored in platelets. Many clinical situations common to critically ill patients lead to increases in PF4, such as diabetes, cardiopulmonary bypass, infections, and renal injury to name a few.

These granules then bind to heparin, which can provoke an antibody-mediated process that leads to platelet activation and destruction in the presence of an antibody to this complex (Fig. 20.12).

The simultaneous activation and destruction of platelets in this process is important, as it can predispose to both thrombocytopenia with associated risks of bleeding, as well as inflammation and thrombosis associated with platelet activation.

The diagnosis of HIT involves the presence of high levels of PF4 as well as the presence of an antibody to this complex.

The diagnosis of HIT can be difficult due to the many alternate causes of thrombocytopenia in patients on ECMO such as sepsis, infections, drugs, and sequestration/consumption by the ECMO circuit.

Heparin Resistance

Heparin resistance refers to the phenomenon of diminished effect of heparin due to relative deficiency of antithrombin. Since the effect of heparin is dependent on the presence of preexisting antithrombin, lower levels of circulating antithrombin will translate into less anticoagulation

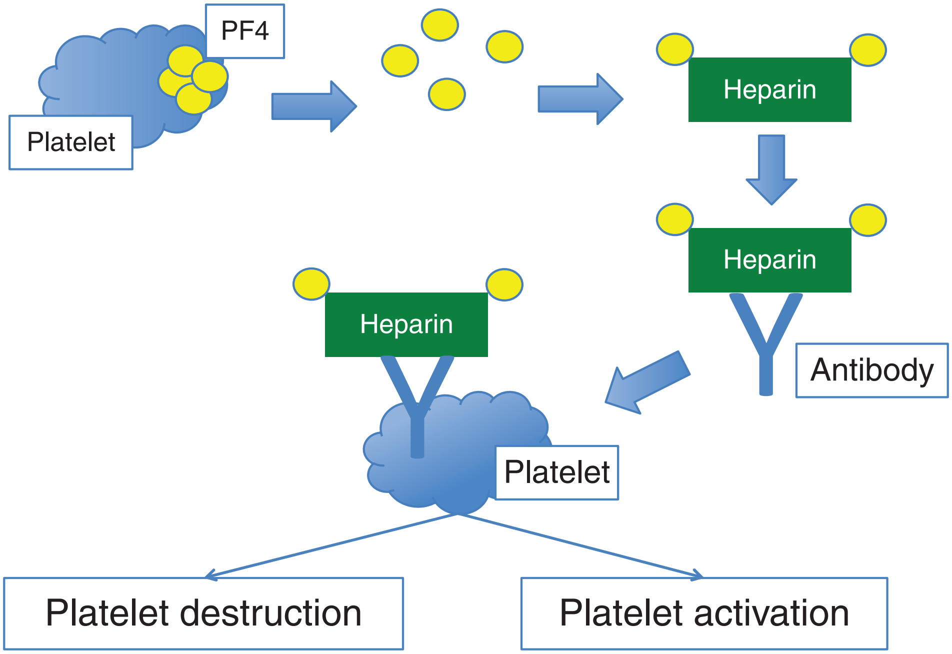

FIG. 20.12 Mechanism of heparin induced thrombocytopenia

activity of heparin. This will manifest itself as escalating doses of heparin with minimal change in PTT.

What can be done if this is the case? Some programs, especially pediatric programs, elect to replete antithrombin. Often however, the answer is simply giving more heparin. This will usually improve the situation, and is a reminder that heparin can be dosed in a weight-based manner – for patients with larger body mass, a high total amount of heparin may be needed to achieve a therapeutic endpoint.

For those patients in whom escalating doses are not affecting coagulation parameters (PTT/anti-Xa levels), where repletion of antithrombin is not feasible or desired, other agents may need to be considered.

Difficulty Maintaining a Therapeutic Window

The final disadvantage of heparin may be difficulty in maintaining a therapeutic window. As heparin's effect is dependent on relative degree of antithrombin, this may translate into variability in PTT/anti-Xa levels. This may be very clinically relevant in patients with a high risk of thrombosis and bleeding, requiring a very narrow therapeutic window to be maintained.

DIRECT THROMBIN INHIBITORS: A POTENTIAL ALTERNATE FOR ANTICOAGULATION?

What can be done in cases of heparin resistance, heparin induced thrombocytopenia, or difficulty maintaining a therapeutic window, where heparin use may be contraindicated or less ideal? This is where some teams turn to direct thrombin inhibitors (DTIs).

Like heparin, DTIs work through inhibition of thrombin activity, which has the effect of decreasing platelet activation, fibrin generation, and activation of the intrinsic cascade. However, unlike heparin, this inhibition occurs directly, with the drug itself inhibiting thrombin rather than depending on antithrombin to do so.

We can imagine the advantages of this approach. There will be less resistance and variability of drug effect based on the relative degree of antithrombin. This may lead to a much smoother titration with less variability. Additionally, if a diagnosis of HIT is made, DTIs may be needed to anticoagulate and prevent the thrombosis associated with HIT without using heparin (Fig. 20.13).

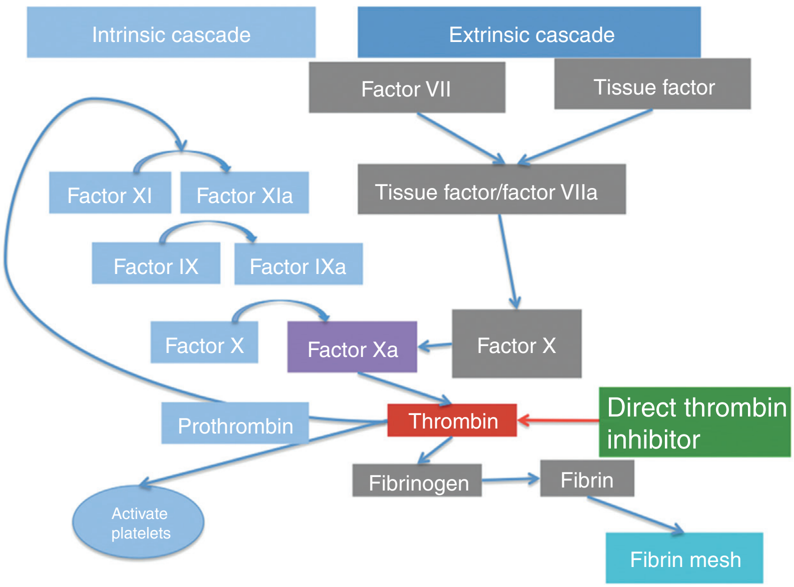

FIG. 20.13 Mechanism of action of direct thrombin inhibitors

However, there are some disadvantages of DTIs that are worth mentioning. The most relavent to many hospitals and programs is cost, as they are significantly more expensive than heparin. Additionally, many practitioners have less experience with DTIs, which may lead to a learning curve when it comes to titration and adjustment. Heparin is still the anticoagulant of choice for cardiopulmonary bypass/catheterization lab, so if repeated trips to the OR/catheterization lab are planned, this may require going back and forth from heparin to DTI, which may lead to error. Finally, unlike heparin, there is no specific reversal agent for DTIs.

DTI Options

There are several DTIs, but the two types that are most commonly used are argartroban and bivalrudin. The primary distinction between them is how they are cleared from the body (Fig. 20.14).

Bivalrudin is primarily cleared renally, while argatroban undergoes hepatic clearance. In general, there are more programs with experience with bivalrudin.

OTHER OPTIONS FOR THROMBOSIS INHIBITION IN ECMO

Remember that at the root, there are two main components of clot – the formation of a thrombin mesh and the aggregation and adherence of a platelet plug. Thus far, we have addressed inhibition of the fibrin mesh and activation of platelets through inhibition of thrombin. However, there may

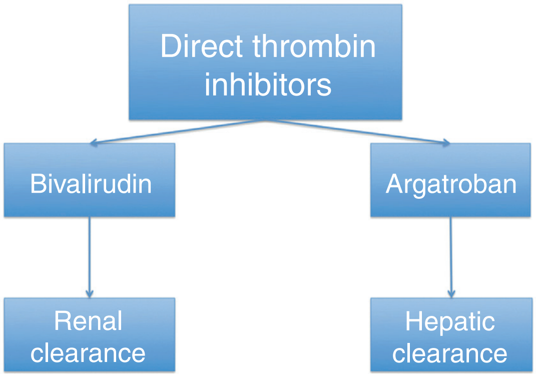

FIG. 20.14 Direct thrombin inhibitors

be the ability for additional mitigation of thrombus formation by addressing the aggregation/adherence of platelets themselves.

This inhibition is often accomplished through the administration of aspirin, a medication with which inhibits the ability of platelets to adhere to the fibrin mesh/vessel wall as well as to aggregate to each other. Addition of aspirin may help to prolong oxygenator life, but should be done with an eye on the additional risks of bleeding that it may add, particularly in patients with low or falling platelet counts.

Let's now turn out attention to bleeding management, and uncover some general principles that we can apply when trying to evaluate the relative risks of clotting and bleeding.

BLEEDING PREVENTION, MITIGATION, AND MANAGEMENT

Why do we have to go into all of this detail? Hopefully at this point, you are starting to develop an appreciation of the system for coagulation and bleeding management. If we incur the risk of ECMO, we need to have a way of addressing and mitigating bleeding and clotting, which can be very common. Thrombosis has been reported to be as high as 25%–30% in the case of clotting, while bleeding has been reported in up to 40%–45% of ECMO cases, with complications ranging from well tolerated to devastating.

Part of mitigating complications on ECMO involves understanding where the patient is on the spectrum between clotting and bleeding (Fig. 20.15).

How can we address and mitigate bleeding for patients on ECMO, especially when the need exists to anticoagulate? As is common to our approach to many problems with ECMO, the answer involves identification of a deficiency, application of a strategy for mitigation, and assessment for response.

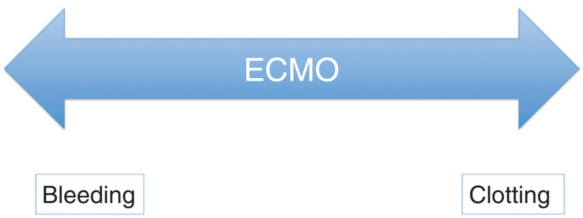

FIG. 20.15 Identifying spectrum of bleeding and thrombosis for patients on ECMO

Identification of Reason for Bleeding

Anticoagulation is a common reason for bleeding but is not the only reason patients bleed while on ECMO. There is consumption of factors and cells by the circuit, both through sequestration by the oxygenator and lysis by the pump, particularly of fibrin and platelets. Critical illness can lead to deficient production of coagulation factors and platelets as well as non-circuit consumption of platelets and factors, such as due to disseminated intravascular coagulation. Finally, turbulence and the effect of the centrifugal pump can cause mechanical disruption of platelet multimers, with an acquired von Willebrand disease effect.

Teasing out this host of causes of bleeding involves careful consideration of the potential causes of bleeding and coagulopathy, with a structured approach for working up bleeding. The typical lab battery can include complete blood count, coagulation factors, fibrinogen, as well as an assessment of the strength of blood clots, such as thromboelastography (TEG). This last test carries the advantage of assessing in real time the ability of the blood to clot and at which point in the clotting cascade the deficiency exists.

Prevention and Mitigation of Bleeding

The most important way to mitigate the risks of bleeding is prevention of clinical situations that may lead to bleeding and prompt intervention when bleeding exists. In general, any unnecessary invasive procedure while on ECMO should be deferred if clinically feasible. As an example, monitoring a small apical pneumothorax may be more prudent than placing a chest tube if the addition of the chest tube will add little to the overall clinical course and will risk massive bleeding. Even procedures considered to be relatively benign such as arterial/venous access, nasogastric tubes, or urinary catheterization can lead to massive hemorrhage and should be deferred to clinicians with adequate experience whenever possible.

Additionally, prompt intervention to bleeding can be prudent and it may allow for abatement of the bleeding before hemorrhage gets out of hand. This can include endoscopy/angiography as needed for GI bleed, return to the OR for surgical site bleeding, and addressing any bleeding around the cannulas with suturing as indicated.

Addressing Bleeding Due to Coagulopathy

Bleeding in the setting of coagulopathy, whether as the result of anticoagulation administered or due to an underlying condition, should prompt assessment of the underlying issue. What follows are several steps that may be addressed to improve the overall ability for forming clot.

1. **Replete products if low:** Sometimes the most prudent intervention is the most apparent. If platelets are low, consider replacing. If PT/INR is elevated, fresh frozen plasma may be of benefit. If fibrinogen is low, cryoprecipitate may help to replete. There is no evidence of the benefits of replacing factors if the levels are not low, and blood product administration carries its own risks and adverse effects.
2. **Guided product repletion based on clot strength:** The advantage of TEG is that it can give assessment of how the blood is clotting which can guide how factors and products can be administered. This may allow for a more targeted approach than just administering all products in the presence of bleeding.
3. **Decrease or hold anticoagulation:** Sometimes just lowering the anticoagulation target or dosage is enough to improve bleeding. This will be a risk-benefit assessment of the potential for further thrombosis/clotting versus bleeding.
4. **Address circuit source of coagulopathy:** If the circuit itself is suspected to be the source of the coagulopathy, due to an older oxygenator consuming of fibrin/platelets, consideration can be

given to changing out the oxygenator. This decision will have to consider risks and benefits of an oxygenator change out.

Consideration can also be given to improving the clot strength while on anticoagulation as detailed later.

IMPROVING CLOT STRENGTH WHILE ON ANTICOAGULATION

As you will recall, Step 6 of our clot regulation process involved breaking down the clot once bleeding subsides. This is accomplished by the activation of plasminogen to plasmin, which breaks down the fibrin mesh into fibrin split products (Fig. 20.16).

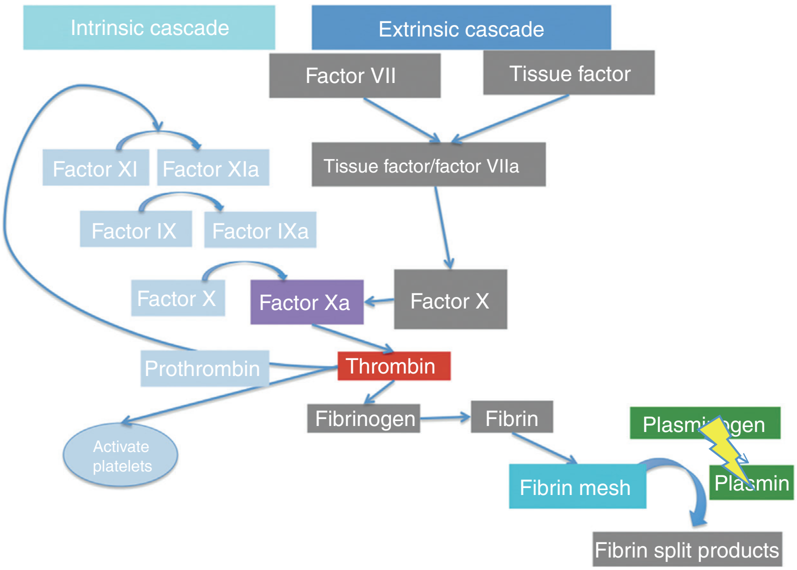

FIG. 20.16 Inhibition of plasminogen activation

Drugs such as transexemic acid and aminocaproic acid inhibit this activation of plasminogen to plasmin. Therefore, the process of fibrinolysis is inhibited. The result is that clots that are already formed are better stabilized and less likely to break down. This strategy can carry the additional advantage of being combined with anticoagulation, with the rationale being that new clots are less likely to form but clots that are formed are less likely to breakdown. This can allow for mitigation of both clotting and bleeding for patients that are at risk for both.

PUTTING IT TOGETHER

Anticoagulation and bleeding management on ECMO can be very challenging. Besides involving very complex systems and pathways in the body, there is unfortunately not a lot of evidence or consensus to guide practice. Indeed, there exists a large variation in how anticoagulation is used, measured, titrated, and held.

It is with an awareness of the paucity of evidence and variation in experience that the concepts in this chapter are presented. Hopefully, the concepts discussed allow for an initial approach to better understand how bleeding and clotting are regulated in the body.

Undoubtedly, more advances are coming in the future – as technology related to the design of the pump, oxygenator, and cannulas continues to improve, this may obviate or even eliminate the need for anticoagulation. In the interim, a systematic approach for evaluating your patient on anticoagulation and intervening when appropriate is likely to serve you well.

SUGGESTED READING

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